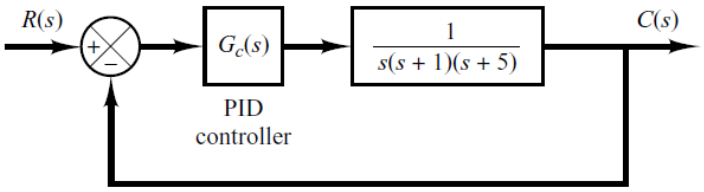
System Dynamics Assignment 4 Solution

1. Consider the control system shown in the figure below in which a PID controller is used to control the system.

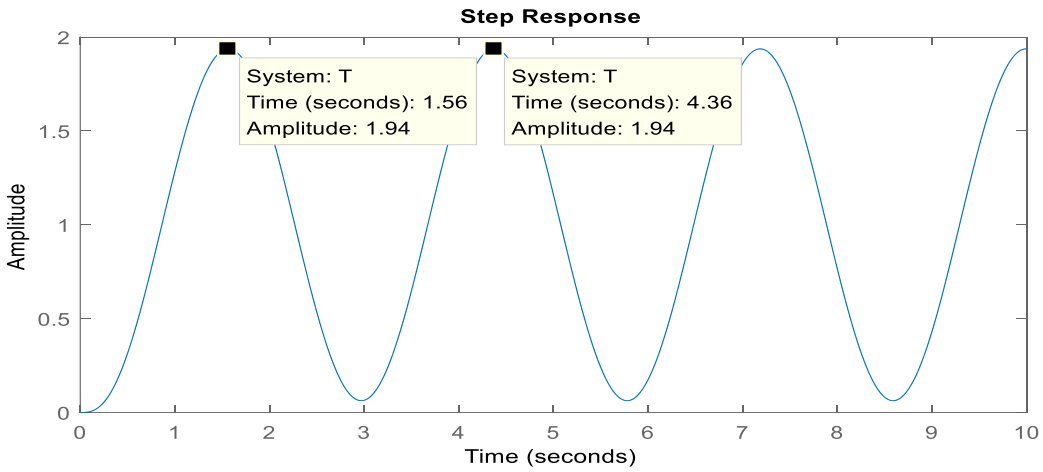


(a) Write the transfer function of the PID controller Gc(s)

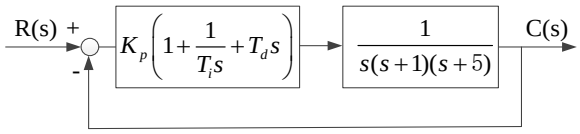
(b) Determine the parameters of the PID controller using Ziegler-nichols method. (Hint: use Matlab simulation to find the critical gain K_{cr} and oscillation period P_{cr} when sustained oscillations are observed.)

Solution: To tune the PID parameter, let $G_c(s) = K_p$

Increase K_p until the sustained oscillations appear in the step response. It is found that at K_p =30 the system will be marginally stable with a oscillatory period of 2.8s. Therefore the critical value K_{cr} = 30.



Therefore, the closed-loop system with a PID controller will be

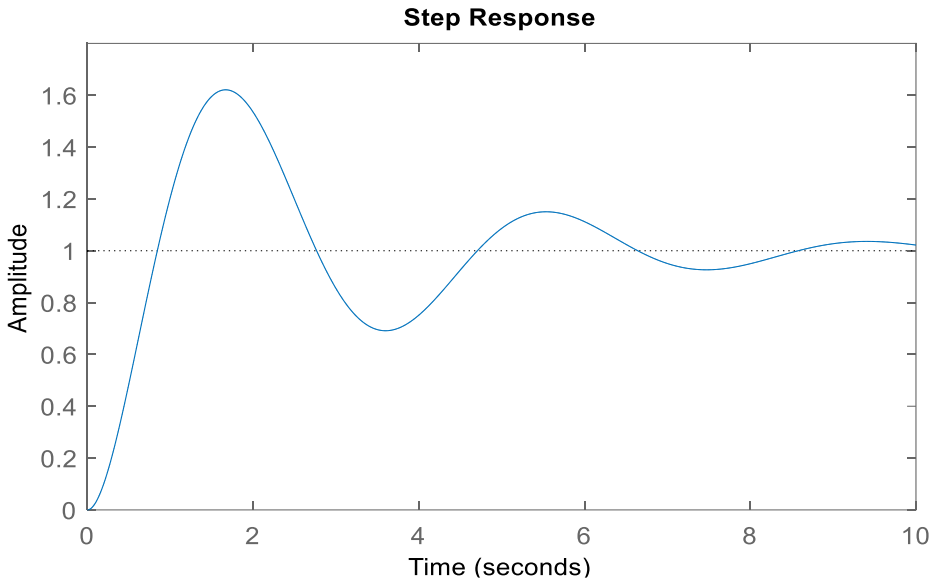


$$\frac{C(s)}{R(s)} = \frac{K_p}{s(s+1)(s+5)+K_p}$$

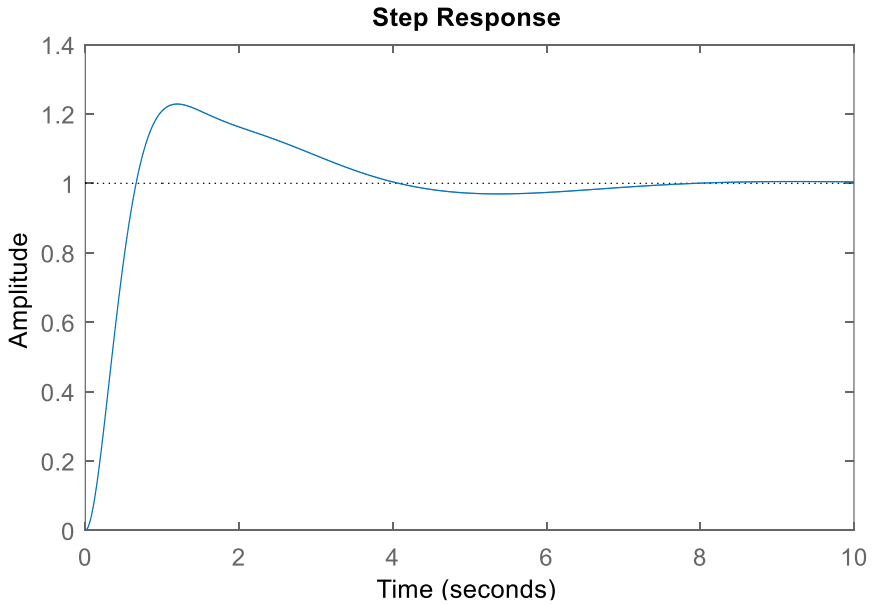
With the second method of Ziegler-Nichols, the PID gains can be obtained as

$$K_p = 0.6K_{cr} = 18$$
$$T_i = 0.5P_{cr} = 1.4$$
$$T_d = 0.125P_{cr} = 0.35$$

With the tuned parameters, the step response can be simulated (shown below)

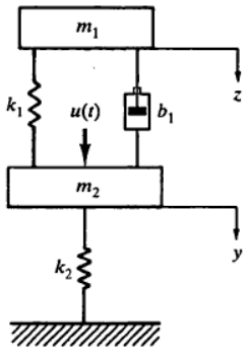


The control performance can be further improved by fine tuning of PID parameters, which leads to



Step response for K_p=18, T_i = 1.4, T_d = 0.85

2. Consider the mechanical system shown in the figure:



1) Obtain a state-space representation with a state variable $\mathbf{x} = [y, \dot{y}, z, \dot{z}]^{\mathrm{T}}$ and the output $\mathbf{y} = [y, z]^{\mathrm{T}}$
2) Given $m_1 = m_2 = 1$, $k_1 = k_2 = 1$, $b_1 = 0.5$, check if the system is completely state controllable and observable.
3) Design a controller such that the system has desired poles at $-\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$, -5,-10

Solution: 1) Assuming y and z are measured from the equilibrium, we have

$$\mathrm{m}_1\text{: } m_1\ddot{z} = -k_1(z-y) - b_1(\dot{z}-\dot{y})$$

$$\mathrm{m}_2\text{: } m_2\ddot{y} = k_1(z-y) + b_1(\dot{z}-\dot{y}) + u - k_2 y$$

which can be rewritten as:

$$m_2\ddot{y} = -(k_1+k_2)y - b_1\dot{y} + k_1 z + b_1\dot{z} + u$$
$$m_1\ddot{z} = k_1 y + b_1\dot{y} - k_1 z - b_1\dot{z}$$

With the state variable $\mathbf{x} = [y, \dot{y}, z, \dot{z}]^{\mathrm{T}}$

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{y} \\ \ddot{y} \\ \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{(k_1+k_2)}{m_2} & -\frac{b_1}{m_2} & \frac{k_1}{m_2} & \frac{b_1}{m_2} \\ 0 & 0 & 0 & 1 \\ \frac{k_1}{m_1} & \frac{b_1}{m_1} & -\frac{k_1}{m_1} & -\frac{b_1}{m_1} \end{bmatrix} \begin{bmatrix} y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m_2} \\ 0 \\ 0 \end{bmatrix} u$$

$$\mathbf{y} = \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x}$$

2) Given $m_1 = m_2 = 1$, $k_1 = k_2 = 1$, $b_1 = 0.5$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -2 & -0.5 & 1 & 0.5 \\ 0 & 0 & 0 & 1 \\ 1 & 0.5 & -1 & -0.5 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = 0$$

The controllability matrix

$$\mathbf{M}_{\mathrm{C}}=\left[\mathbf{B},\ \mathbf{AB},\ \mathbf{AAB},\ \mathbf{AAAB}\right]=\begin{vmatrix}0 & 1 & -0.5 & -1.5\\ 1 & -0.5 & -0.5 & 2.5\\ 0 & 0 & 0.5 & 0.5\\ 0 & 0.5 & 0.5 & -2\end{vmatrix},\ \det\left(\mathbf{M}_{\mathrm{C}}\right)=1,\mathrm{rank}\left(\mathbf{M}_{\mathrm{C}}\right)=4$$

The observability matrix

$$\mathbf{M}_{\mathrm{O}}=\begin{bmatrix}\mathbf{C}\\ \mathbf{CA}\\ \mathbf{CAA}\\ \mathbf{CAAA}\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ -2 & -0.5 & 1 & 0.5\\ 1 & 0.5 & -1 & -0.5\\ 1.5 & -1.5 & -1 & 0.5\\ -1.5 & 0.5 & 1 & -0.5\end{bmatrix},\ \mathrm{rank}\left(\mathbf{M}_{\mathrm{O}}\right)=4$$

Therefore, the system is completely state controllable and also observable.

3) Design a controller with pole placement method, u = **-Kx,** where **K** = [k_a,k_b,k_c,k_d];

The characteristic equation of system with state feedback is

$$\left|sI-A+BK\right|=\left|\begin{bmatrix}s & 0 & 0 & 0\\ 0 & s & 0 & 0\\ 0 & 0 & s & 0\\ 0 & 0 & 0 & s\end{bmatrix}-\begin{bmatrix}0 & 1 & 0 & 0\\ -2 & -0.5 & 1 & 0.5\\ 0 & 0 & 0 & 1\\ 1 & 0.5 & -1 & -0.5\end{bmatrix}+\begin{bmatrix}0 & 0 & 0 & 0\\ k_a & k_b & k_c & k_d\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}\right|$$

$$=\left|\begin{bmatrix}s & -1 & 0 & 0\\ 2+k_a & s+0.5+k_b & -1+k_c & -0.5+k_d\\ 0 & 0 & s & -1\\ -1 & -0.5 & 1 & s+0.5\end{bmatrix}\right|$$

$$=s^4+\left(1+k_b\right)s^3+\left(3+k_a+\frac{k_b}{2}+\frac{k_d}{2}\right)s^2+\left(\frac{1}{2}+\frac{k_a}{2}+k_b+\frac{k_c}{2}+k_d\right)s+k_a+k_c+1$$

which should be the same as the characteristic equation derived using the desired poles:

$$=(s+\frac{1}{2}+j\frac{\sqrt{3}}{2})(s+\frac{1}{2}-j\frac{\sqrt{3}}{2})(s+5)(s+10)$$
$$=s^4+16s^3+66s^2+65s+50$$

K = [k_a,k_b,k_c,k_d] = [43, 15, 6, 25], and the controller is

$$u=-43x_1-15x_2-6x_3-25x_4=-43y-15\dot{y}-6z-25\dot{z}$$